

Day 1: Overview of Hardware Components

Task 1 : Draw a labeled diagram of a desktop PC, clearly marking the motherboard, CPU, RAM, GPU, and both HDD and SSD storage components.



Task 2 : Create a comparison table showing the main differences between HDD and SSD storage in terms of speed, durability, and typical use cases.

Feature	HDD (Hard Disk Drive)	SSD (Solid State Drive)
Speed	Slower	Much faster
Durability	Less durable because it has moving parts	More durable and shock-resistant
Storage Capacity	Usually offers large capacity at lower cost	Usually more expensive per GB
Typical Use	Desktop PCs, bulk storage, backups	Laptops, gaming PCs, operating systems, applications
Noise	Can make some noise while operating	Silent

Power Consumption	Higher	Lower
Best For	Large files and budget storage	Fast booting, gaming, and quick application loading

Task 3 : Write a short explanation (2-3 sentences each) describing the role of the CPU, RAM, and GPU in running a game like PUBG or Free Fire on a PC.

1. CPU:

- The **CPU (Central Processing Unit)** is the brain of the PC and handles game calculations, player movements, physics, and other instructions.
- A good CPU helps the game run smoothly and reduces lag.

2. RAM:

- **RAM (Random Access Memory)** temporarily stores the game data and programs that are currently being used.
- More RAM allows the game to run smoothly while other applications are open in the background.

3. GPU:

- The **GPU (Graphics Processing Unit)** renders the game's graphics, including characters, maps, textures, and visual effects.
- A powerful GPU provides higher FPS and smoother gameplay.

Task 4 : Imagine you want to upgrade your PC to run the latest version of FIFA smoothly. List which hardware components you would consider upgrading and explain why for each.

1. GPU (Graphics Card)

- I would upgrade the GPU because it has a major role in rendering game graphics.
- A better GPU can provide higher FPS and smoother gameplay.

2. CPU (Processor)

- A faster CPU helps process game calculations, player movements, and other game instructions.
- This can reduce lag and improve overall performance.

3. RAM

- I would consider upgrading to at least **16 GB RAM** for smooth gaming and better multitasking.
- It helps the game load and run without slowing down the system.

4. SSD (Storage)

- Upgrading to an SSD can make the game and Windows load much faster.
- It also reduces loading times compared with an HDD.

5. Power Supply (PSU)

- If I install a more powerful GPU, I would check whether the existing PSU can provide enough power.
- If not, I would upgrade to a suitable PSU to keep the PC stable.